

GTU CSE 241 / 501 Quiz

Std Code: 142

Std Number:

2 1 0 1 0 4 0 0 4 0 1 7

```
#ifndef MY_HEADER
#define MY_HEADER

#include <iostream>
#include <stdexcept> //for exceptions

template<class T> //or type name, same thing.
class GTUSet {
public:
    GTUSet() { SetElements(0);
                //or SetElements.resize(0); }

    void add(const T& j) {
        if (isIn(j)) throw std::invalid_argument("");
        else SetElements.push_back(j);
    }

    bool isIn(const T& j) const {
        for (const auto& p : SetElements) {
            if (p == j) return true;
        }
        return false;
    }
private:
    std::vector<T> SetElements;
};

#endif
```

```
int main() {
```

```
    GTUSet<int> g; // could be any type.
```

```
    for (int i = 0; i < 5; i++)
```

```
        g.add(i);
```

```
    if (g.isIn(3)) std::cout << "set has 3" << std::endl;
```

```
    if (g.isIn(4)) { // specific case to catch
```

```
        try { // try catch is obligation in add function.
```

```
            g.add(4);
```

```
        } catch (const std::invalid_argument & e) {
```

```
            std::cerr << "the error was happened" << e.what() << std::endl;
```

```
        }
```

```
    for (int i = 2; i < 6; i++) {
```

```
        try { // just showing try catch again.
```

```
            g.add(i);
```

```
        } catch (const std::invalid_argument & e) {
```

```
            std::cerr << "error !" << e.what() << std::endl;
```

```
        }
```

```
    }
```

```
    return 0;
```

```
}
```